Power Budget

Team Number: 203
Project Name: **EGR 304 Cat-Toy**
Team Member Names: **Stellan, Joseph, Ramon, Samuel**
Version: **1**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	9V Battery	A1604 BK210J	-9V	1	300	300	mA
	Speaker	FR 8 - 8 OHM	(Range)	1	625	625	mA
	Transistor 1	2N3904	+0 - 40V	1	200	200	mA
	Transistor 2	2N3906	+0 - 40V	1	200	200	mA
	Microcontroller	PIC18F57Q43	+1.8-5.5V	1	350	350	mA
	Opamp	MCP6004-I/P	+5 to -5	0	23	0	mA
	5V regulator	L7805ABV-DG	+5V - 35V	1	1500	1500	mA
	Barrel Jack	PJ-102AH	+0 - 24V	1	5000	5000	mA

+5V Power Rail

Component Name	Part Number	Supply	Qty.	Absolute	Total	Unit
Transistor 1	2N3904	+0 - 40V	1	200	200	mA
Transistor 2	2N3906	+0 - 40V	1	200	200	mA
Microcontroller	PIC18F57Q43	+1.8-5.5V	1	350	350	mA
Speaker	FR 8 - 8 OHM	(Range)	1	200	200	mA
				Subtotal	950	mA
				Safety Margin	25%	
				Total Current Required on +5V Rail	1187.5	mA

Regulator

+5V Regulator	L7805ABV-DG	+5V - 35V	1	1500	1500	mA
				Total Remaining Current Available on +5V Rail	312.5	mA

External Power Source 1	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source 1 Selection	Plug-in Wall Supply	930	110VAC	+9V	3000	3000	mA
Power Rails Connected to External Power Source 1	+5V Regulator	LM7805	+5V - 35V	1	1500	1500	mA
	Total Remaining Current Available on External Power Source 1					1500	mA

External Power Source 2	Component Name	Part Number	Supply	Output	Absolute	Total	Unit
Power Source 2 Selection	Battery	A1604 BK210J	+9V	+9V	300	500	mA
Power Rails Connected to External Power Source 2	5V Regulator	L7805ABV-DG	+5V - 35V	1	1500	1500	mA
				Total Remaining Current Available on External Power Source 2		-1000	mA

Calculated Battery Life							
	Component Name	**Part Number**	**Supply**		**Capacity**	**Required**	
	Battery	A1604 BK210J	+9V		614	1500	
					Battery Life	0.40933333	hours

Notes:

External Supply Voltage
If you have multiple units